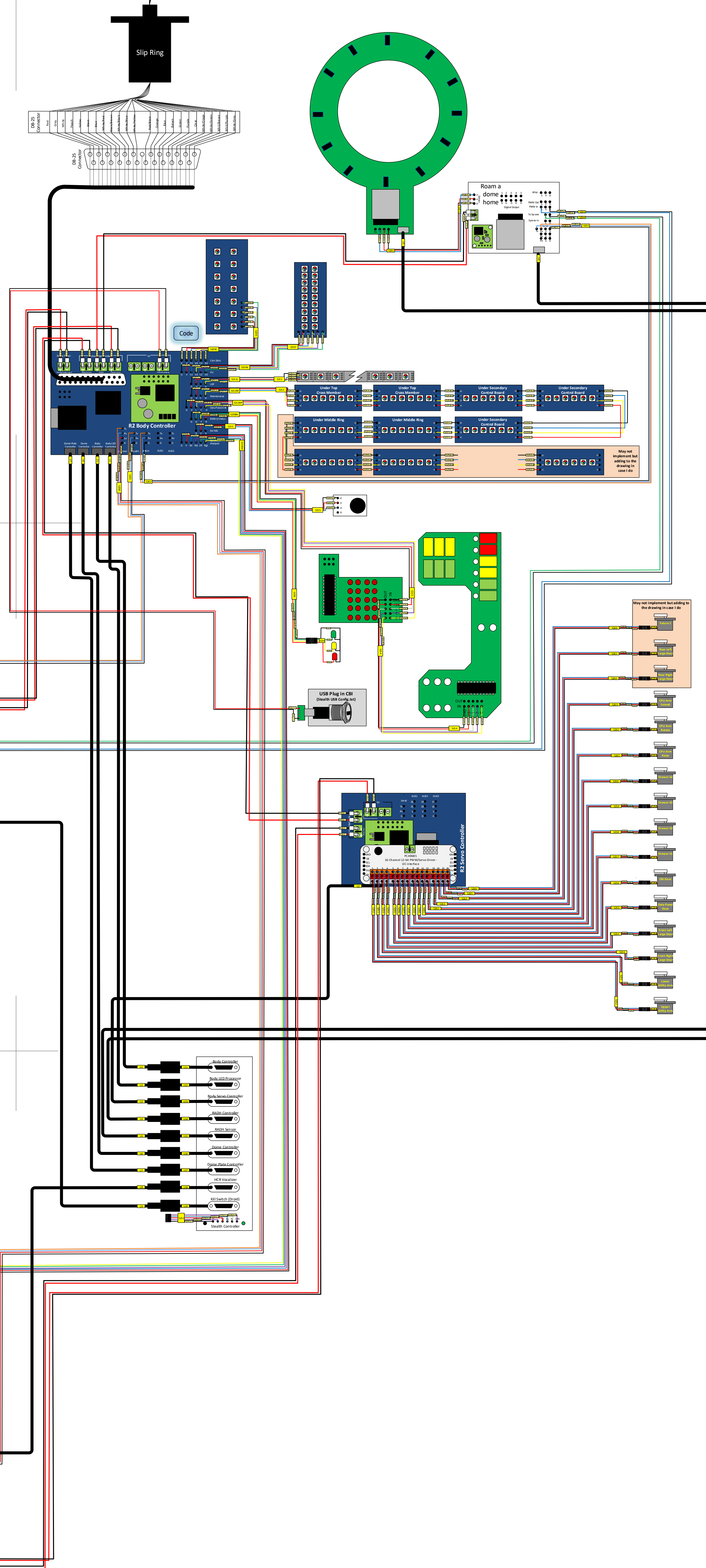
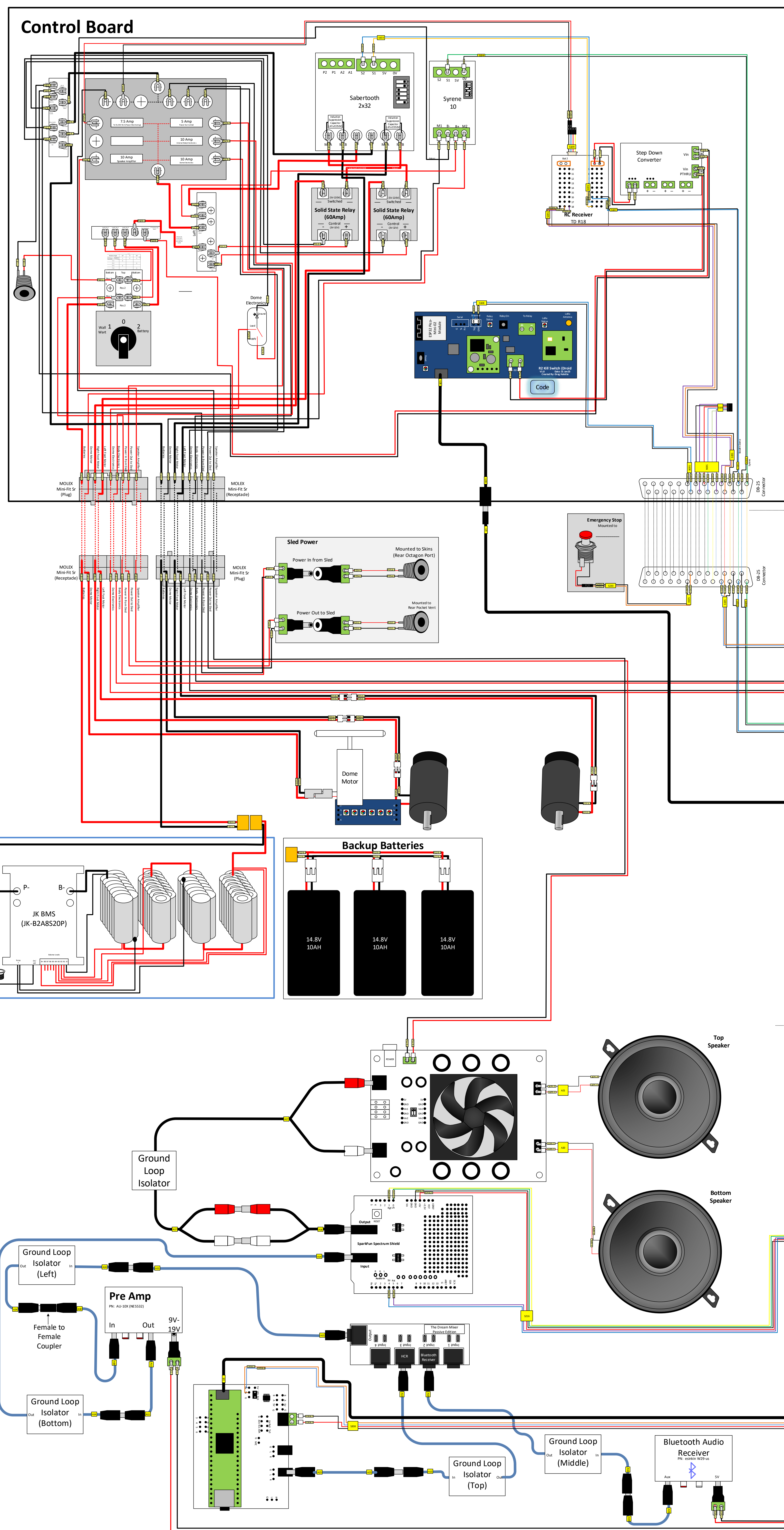
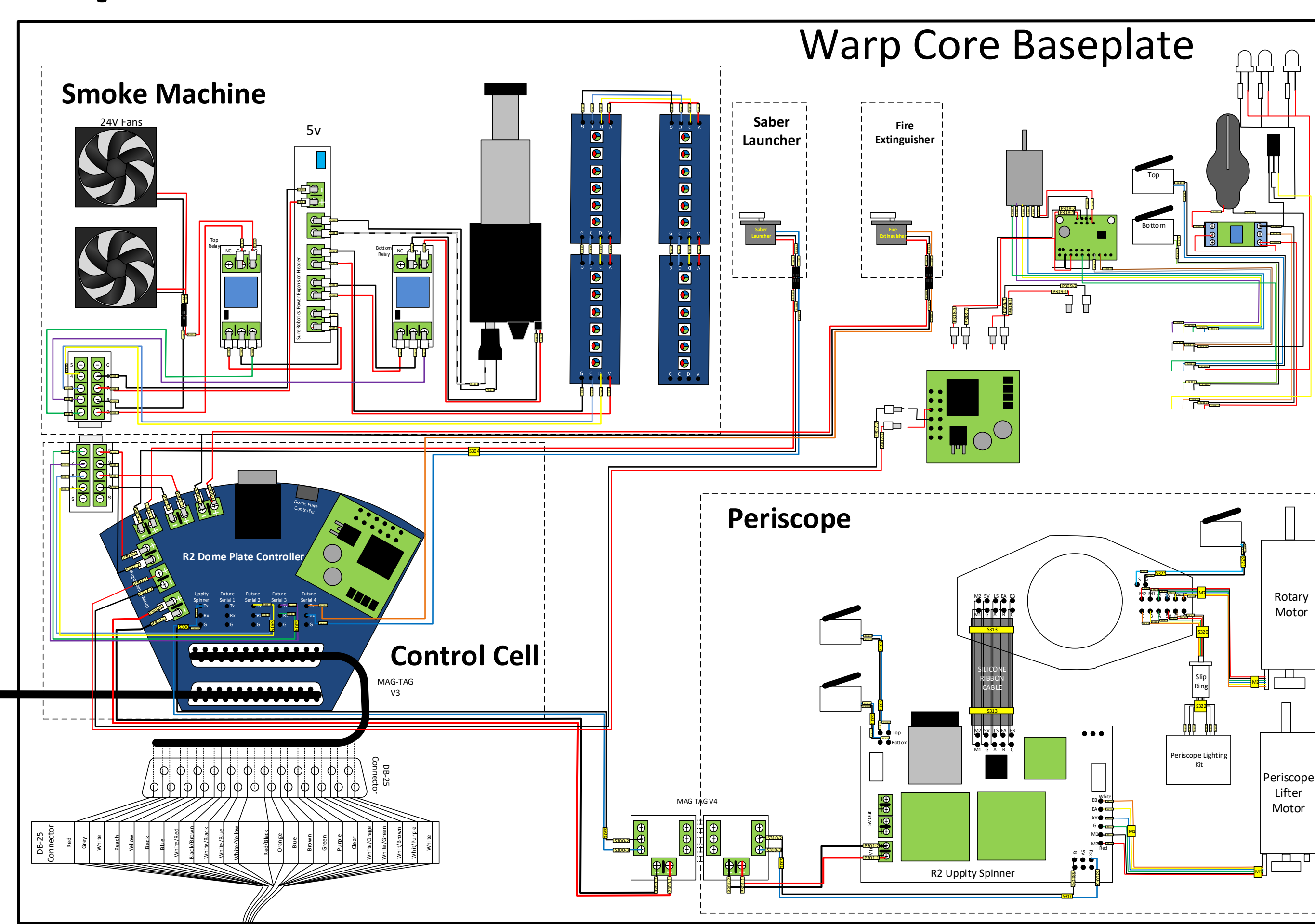
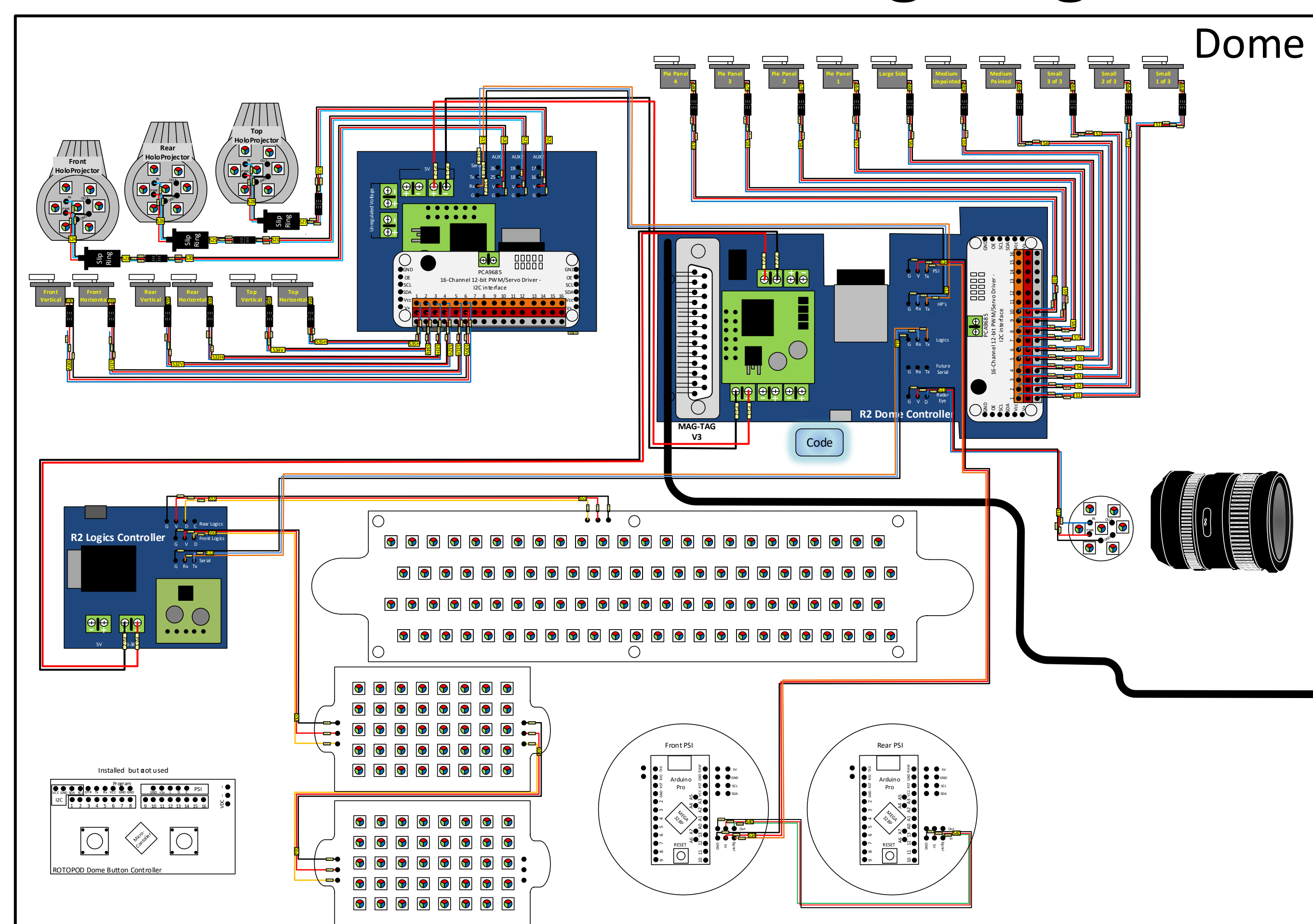
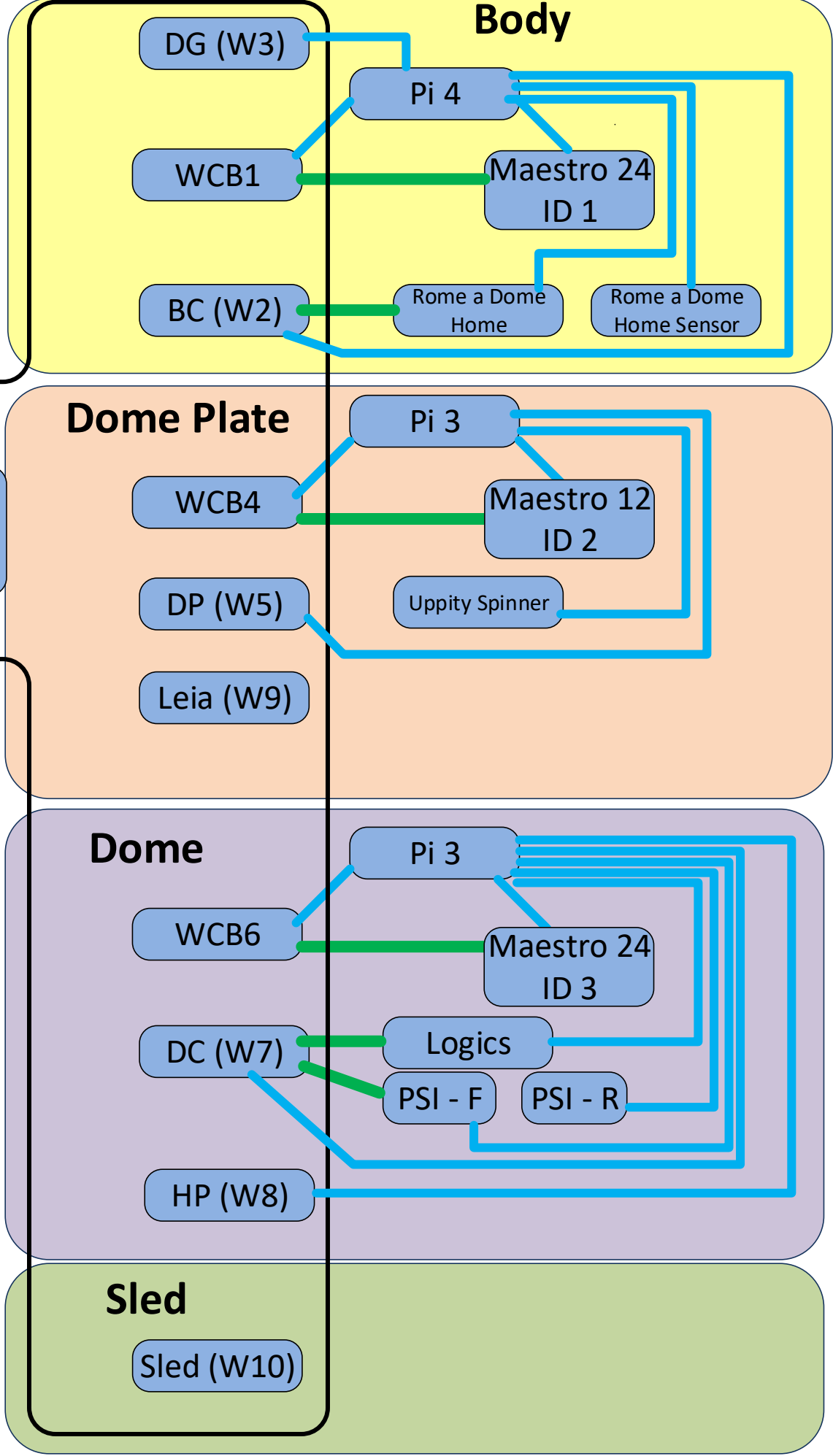


R2D2-Wiring-Diagram--05-Sep-2026-10-31-PM



ESP-NOW

Mesh



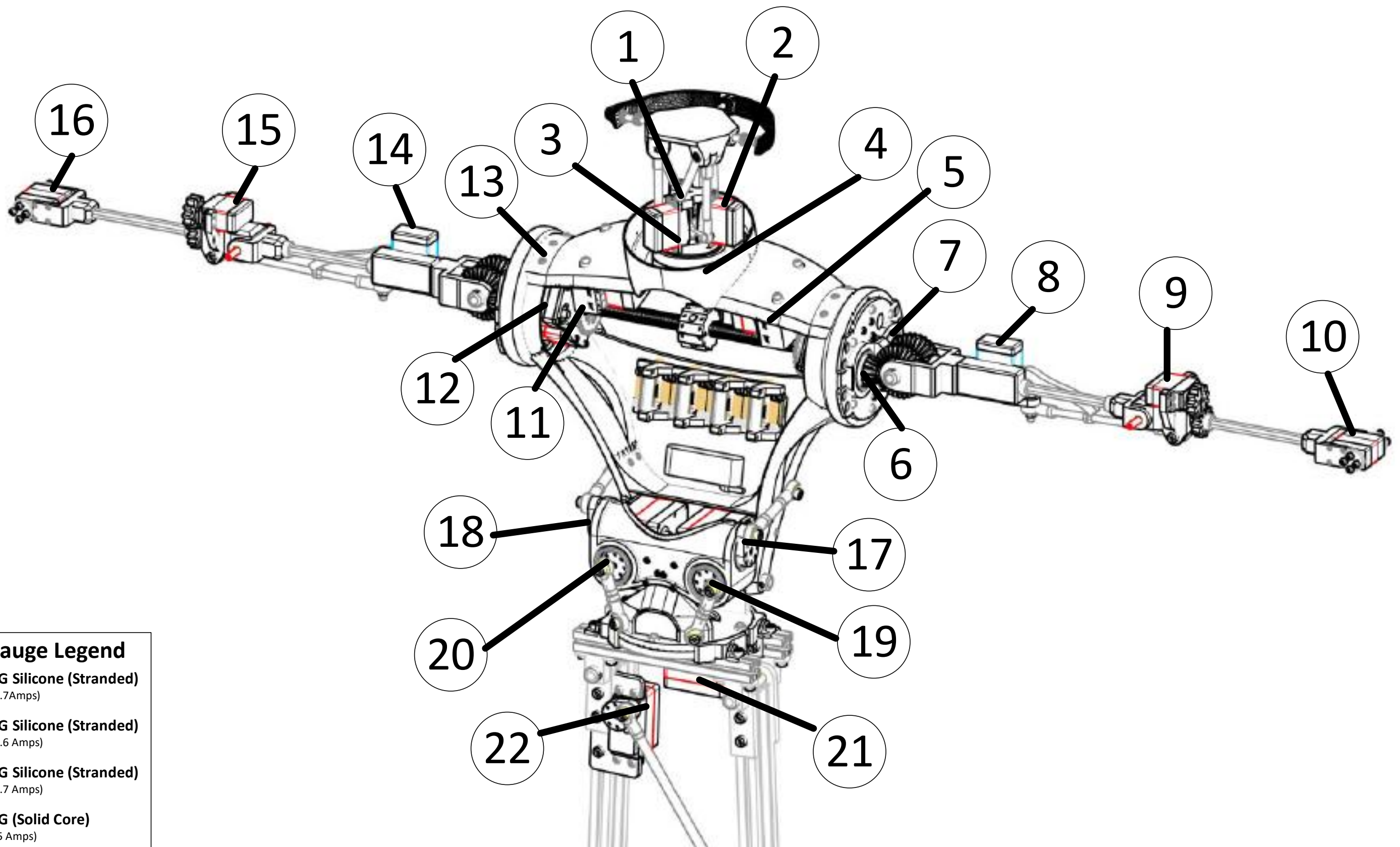
Legend

Serial/UART

USB

SBUS

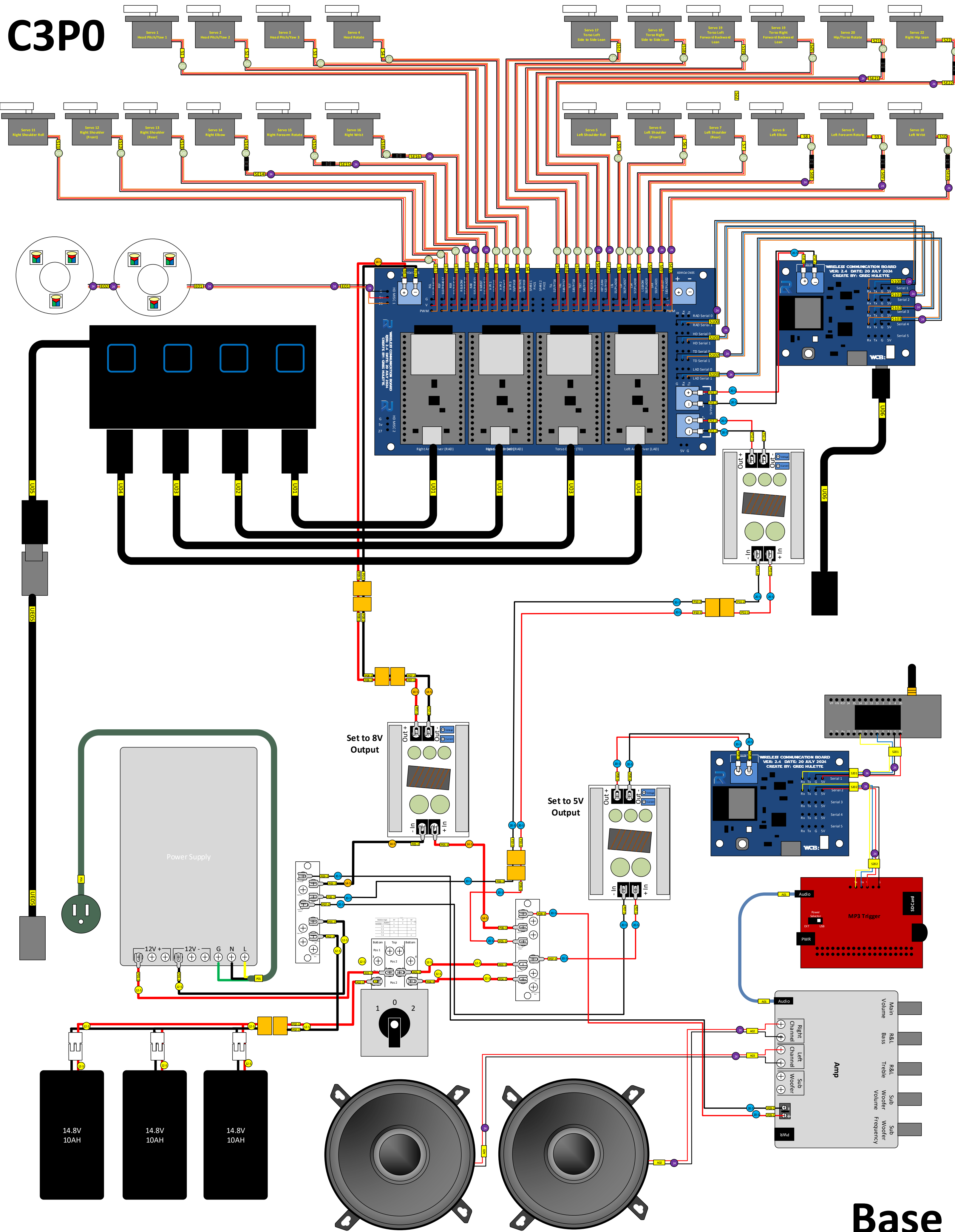
C3P0 Servo Layout



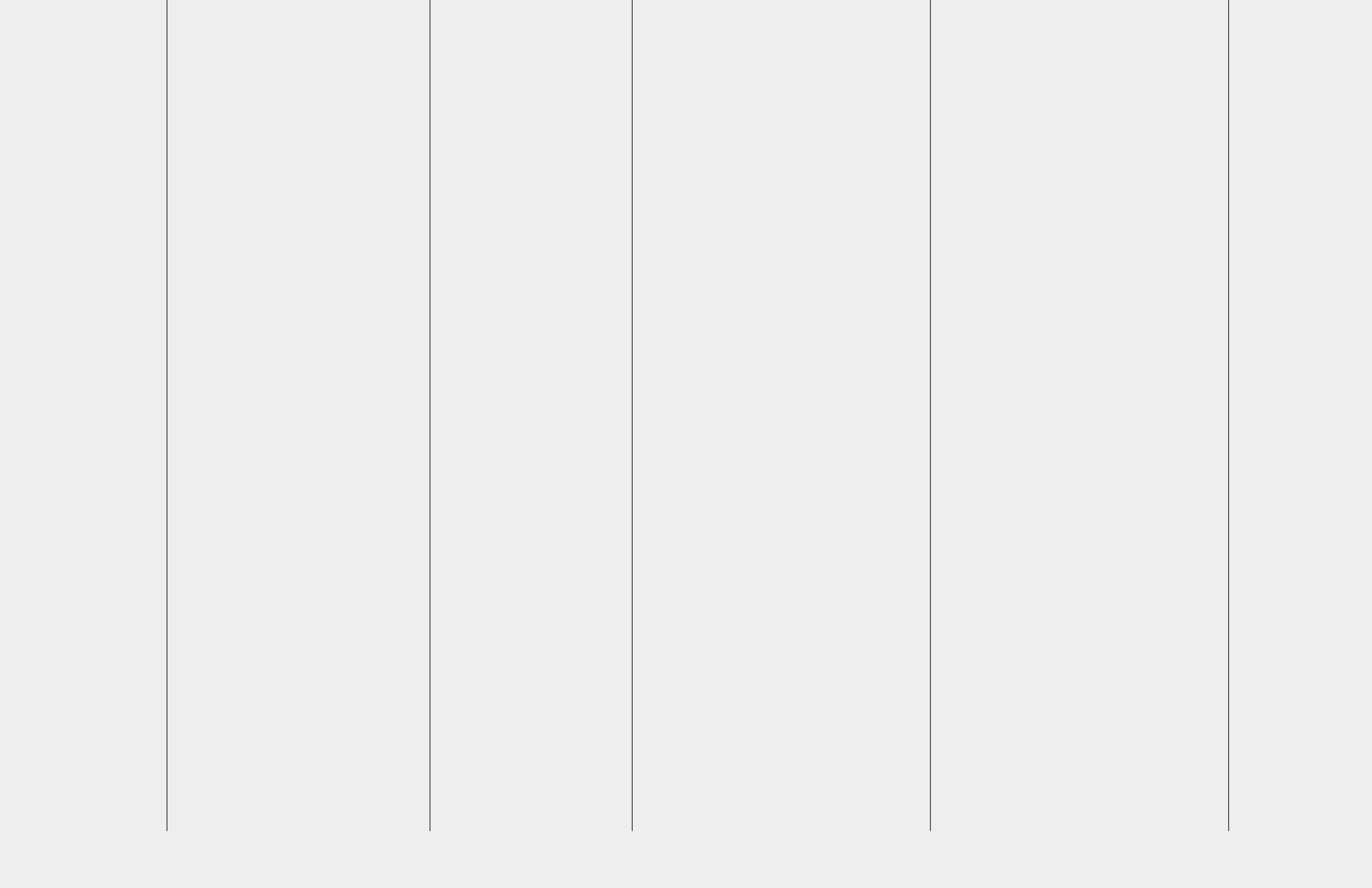
Wire Gauge Legend

12-S	12 AWG Silicone (Stranded)	(Up to 84.7 Amps)
14-S	14 AWG Silicone (Stranded)	(Up to 55.6 Amps)
20-S	20 AWG Silicone (Stranded)	(Up to 13.7 Amps)
24	24 AWG (Solid Core)	(Up to 3.5 Amps)
Str	?? AWG (Stranded)	

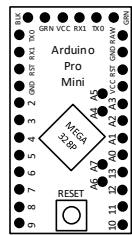
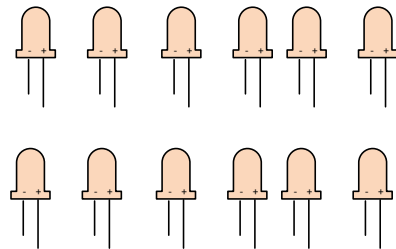
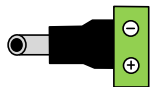
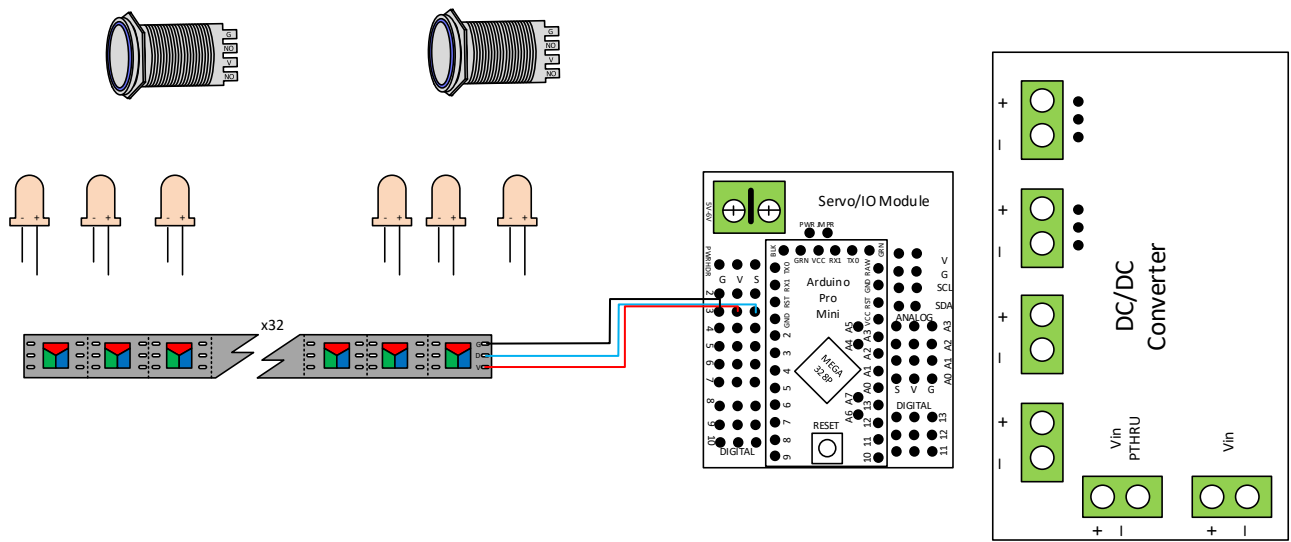
C3P0



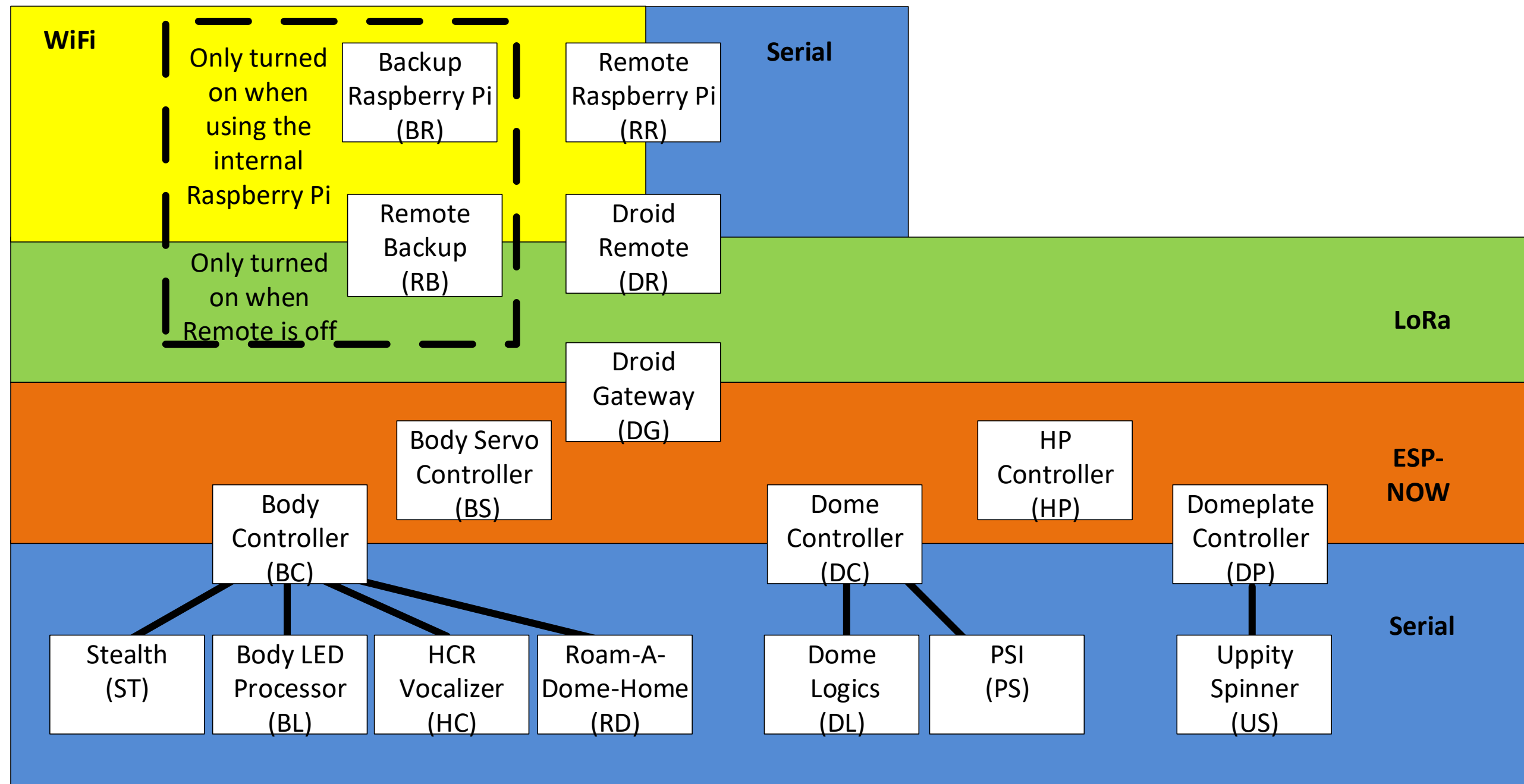
Base



Needs a lot of work



Communication Layers



SF: Maint Lights
Down: Off
Forward: On

SE: Function Select
DOWN

SA: Drive Mode
Down: Standby
Middle: Active
Up: Dome Only

SB: RADH Mode
Rear: Disabled
Middle: Enabled
Up: Enabled

SC: Light Mode
Down: Normal
Middle: All On
Up: All Off

SH:

SG: Slow Drive
Down: Normal
Middle: Slow Mode
Up:

SD: Muse
Down: Muse On
Middle: Muse Off
Up: Muse Off

HCR Vocalizer
Volume

HCR WAV
Volume

Harlem Shake

Panel Wave

Periscope Seq 4

Short Circuit

All Close
Periscope
Down

Wave
Arm

Drawer
Wave

Door
Easing

Fast
Wave

Display

Panel
Wave 2

All Open

Mute HCR

Happy

Mad

Sad

Scared

Fart



SF: Maint Lights
Down: Off
Forward: On

SE: Function Select
MIDDLE

SA: Drive Mode
Down: Standby
Middle: Active
Up: Dome Only

SB: RADH Mode
Rear: Disabled
Middle: Enabled
Up: Enabled

SC: Light Mode
Down: Normal
Middle: All On
Up: All Off

SH:

SG: Slow Drive
Down: Normal
Middle: Slow Mode
Up:

HCR Vocalizer
Volume

HCR WAV
Volume

SD: Muse
Down: Muse On
Middle: Muse Off
Up: Muse Off

Periscope up
and lights on
Periscope
Home

Periscope Rotate
Random
Periscope Seq 10

Flutter

Smoke
Sequence

CPU Arm
Sequence

Long
Dance

Fire
Exting

Arm
Saber

Mute HCR

Dance Star Wars

Leia Message

Star Wars Theme

Vadar Theme

Cantina Song

SF: Maint Lights
Down: Off
Forward: On

SE: Function Select
UP

SA: Drive Mode
Down: Standby
Middle: Active
Up: Dome Only

SB: RADH Mode
Rear: Disabled
Middle: Enabled
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SC: Light Mode
Down: Normal
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SH:

SG: Slow Drive
Down: Normal
Middle: Slow Mode
Up:

HCR Vocalizer
Volume

HCR WAV
Volume

SD: Muse
Down: Muse On
Middle: Muse Off
Up: Muse Off

CPU Arm
Sequence

CPU Arm
Rotate

CPU Arm
Raise

CPU Arm
Lower

CPU Arm
Extend

CPU Arm
Retract

Stow
Saber

Fire
Saber

Mute HCR

Whistle

Staying Alive

SW Quiet

Vader Quiet

Harlem Shake



SF: Maint Lights
Down: Off
Forward: On

SE: Function Select
UP

SA: Drive Mode
Down: Standby
Middle: Active
Up: Dome Only

SB: RADH Mode
Rear: Disabled
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SC: Light Mode
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SH:

SG: Slow Drive
Down: Normal
Middle: Slow Mode
Up:

HCR Vocalizer
Volume

HCR WAV
Volume

SD: Muse
Down: Muse On
Middle: Muse Off
Up: Muse Off

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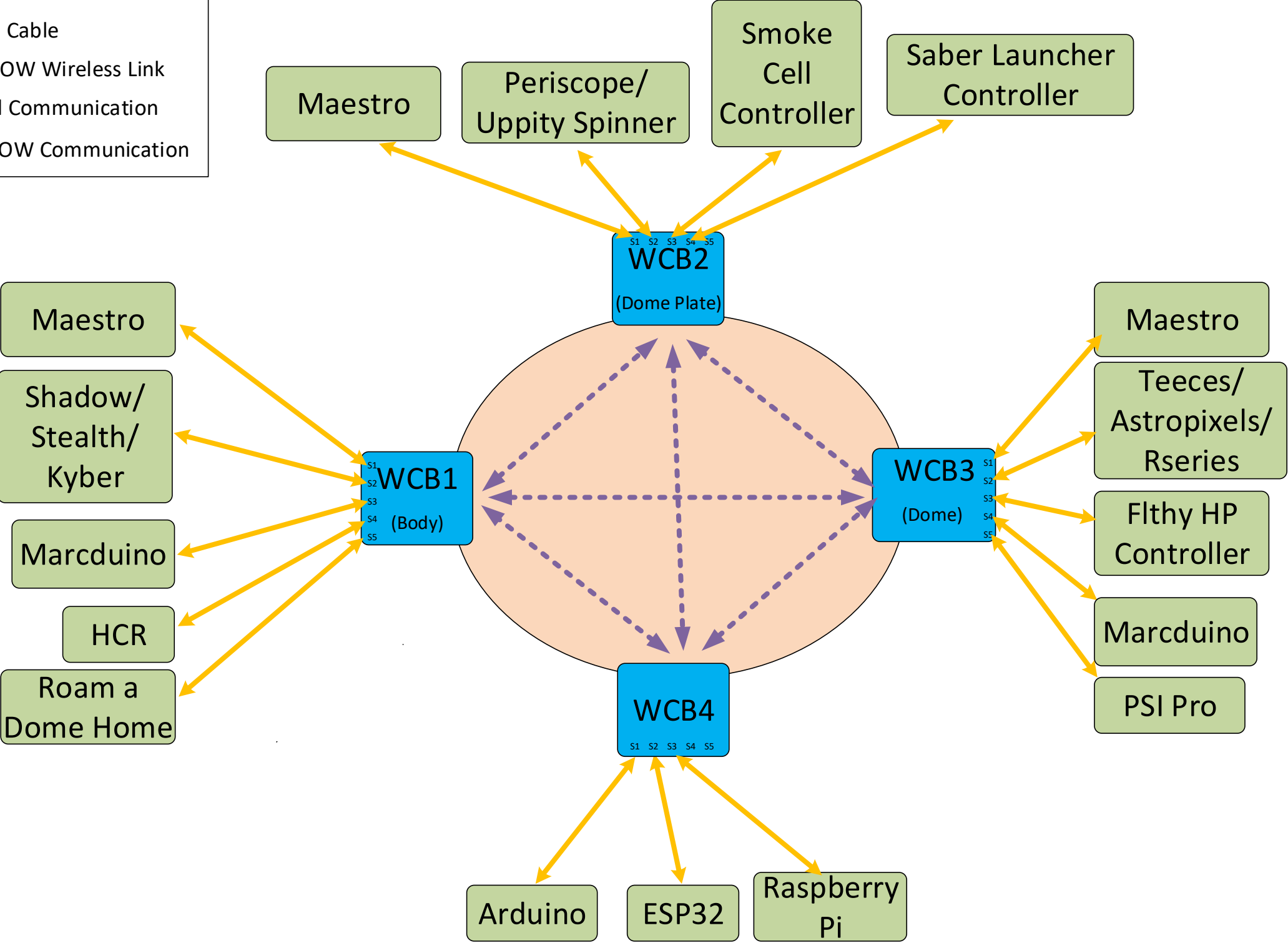
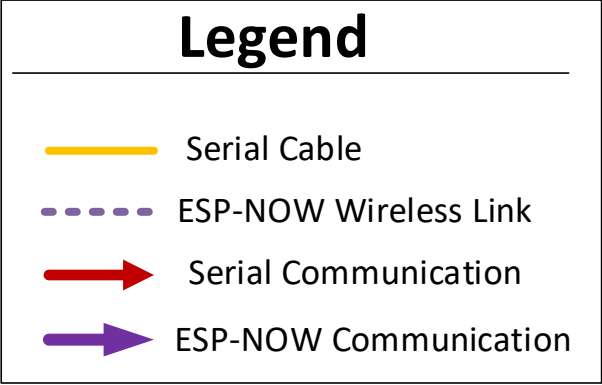
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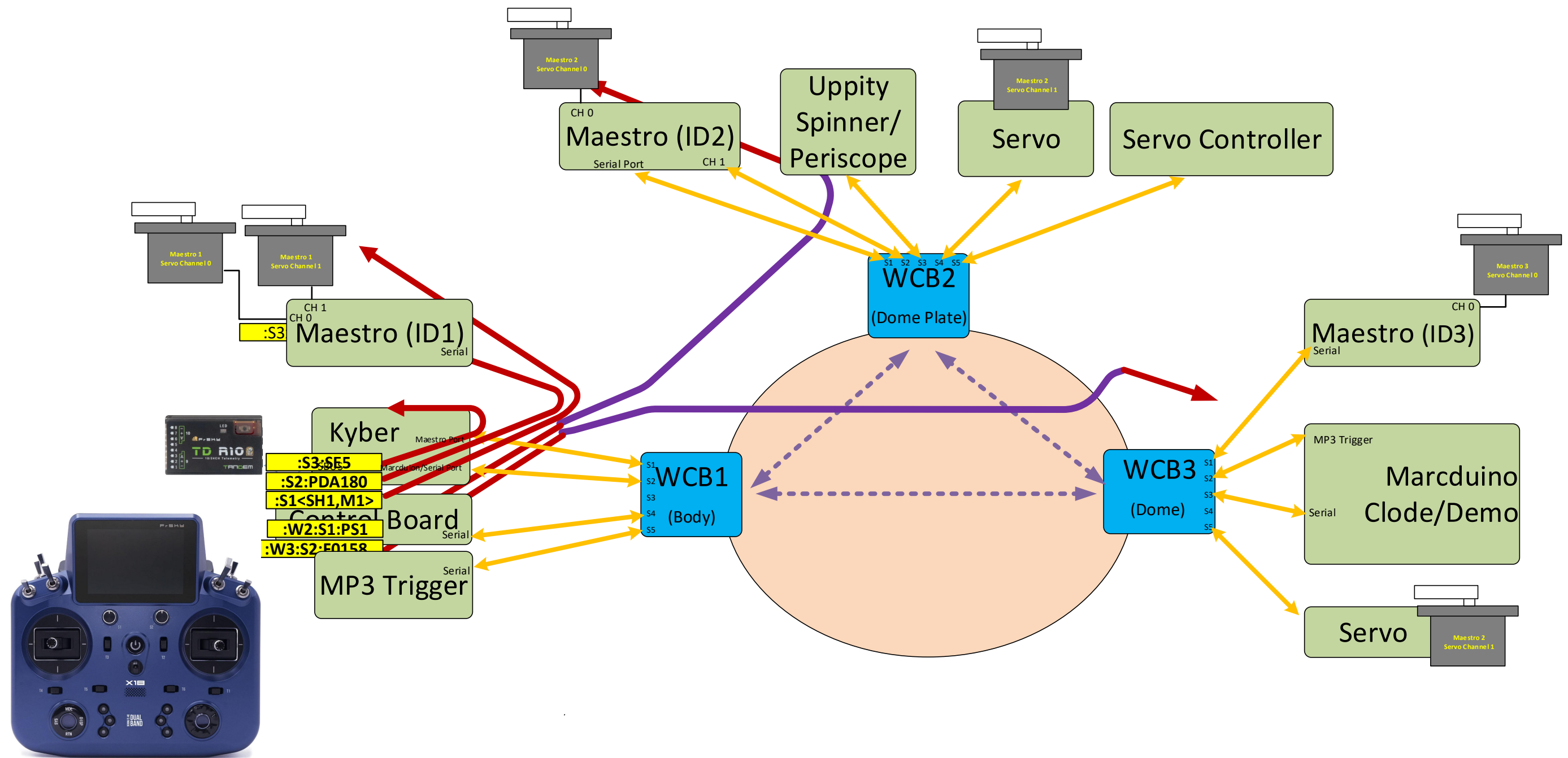
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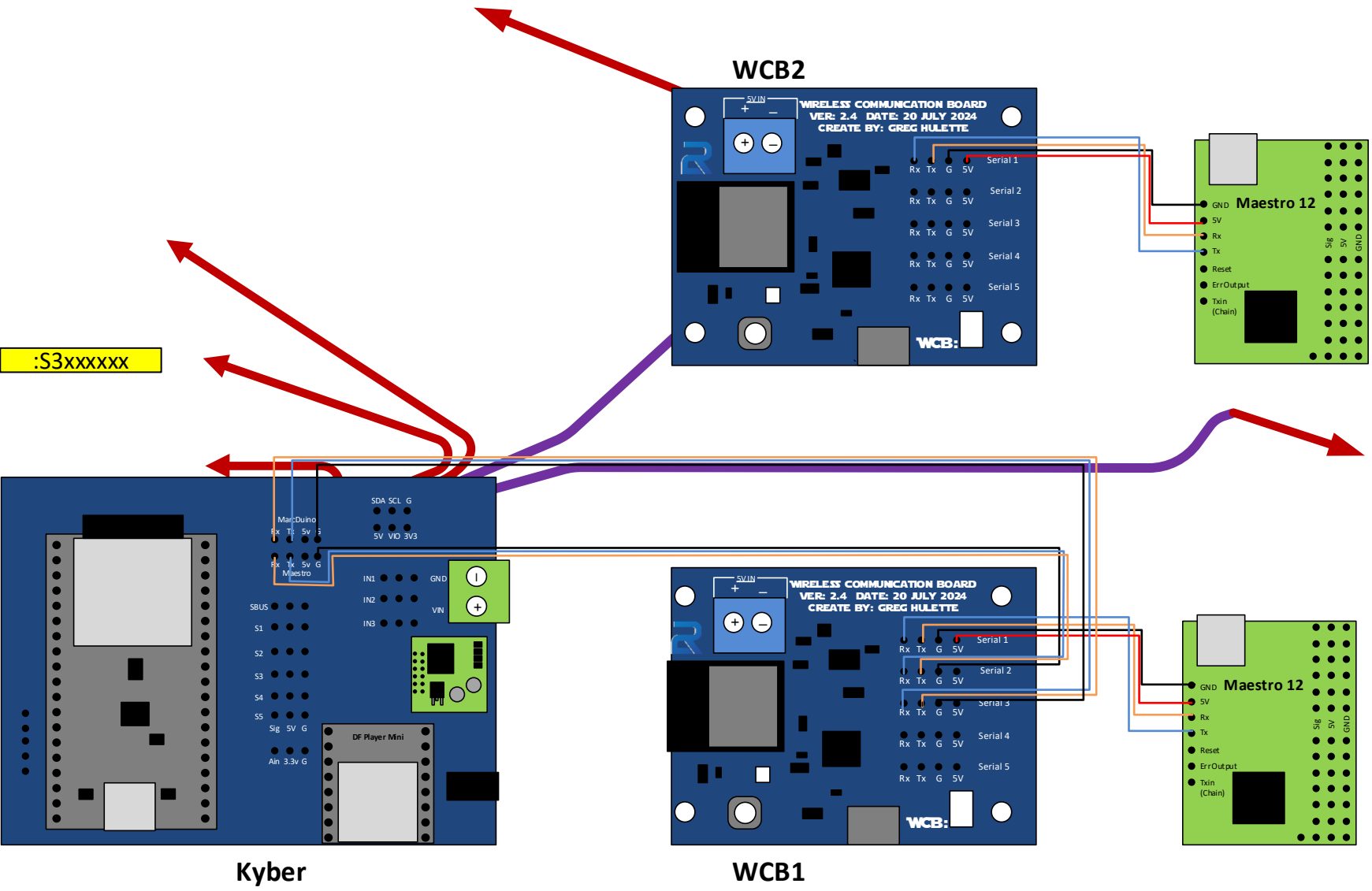
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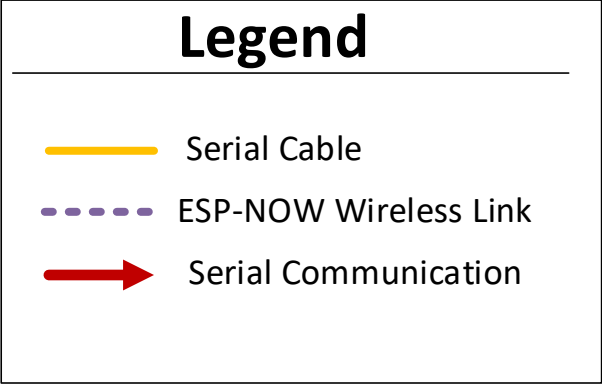
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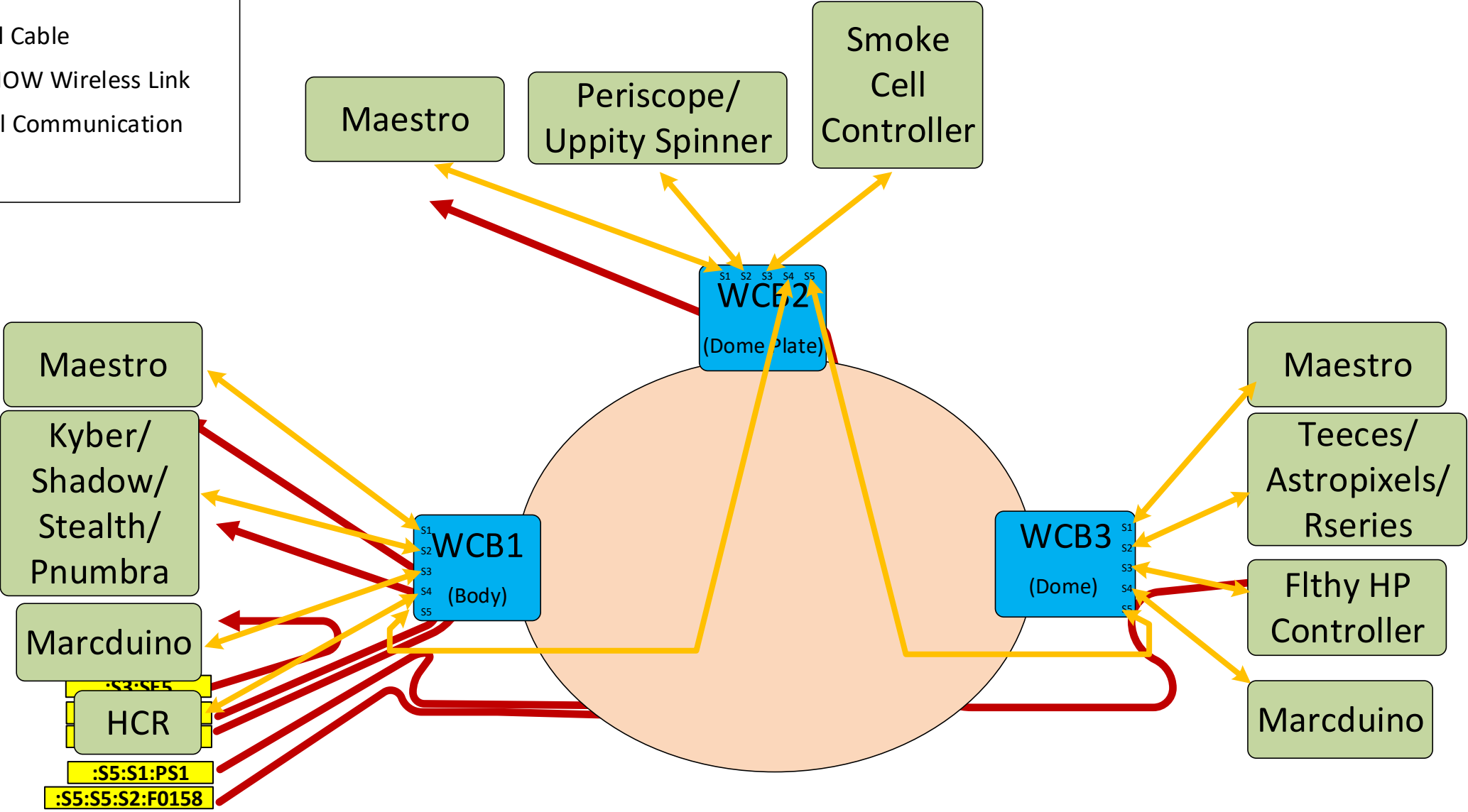




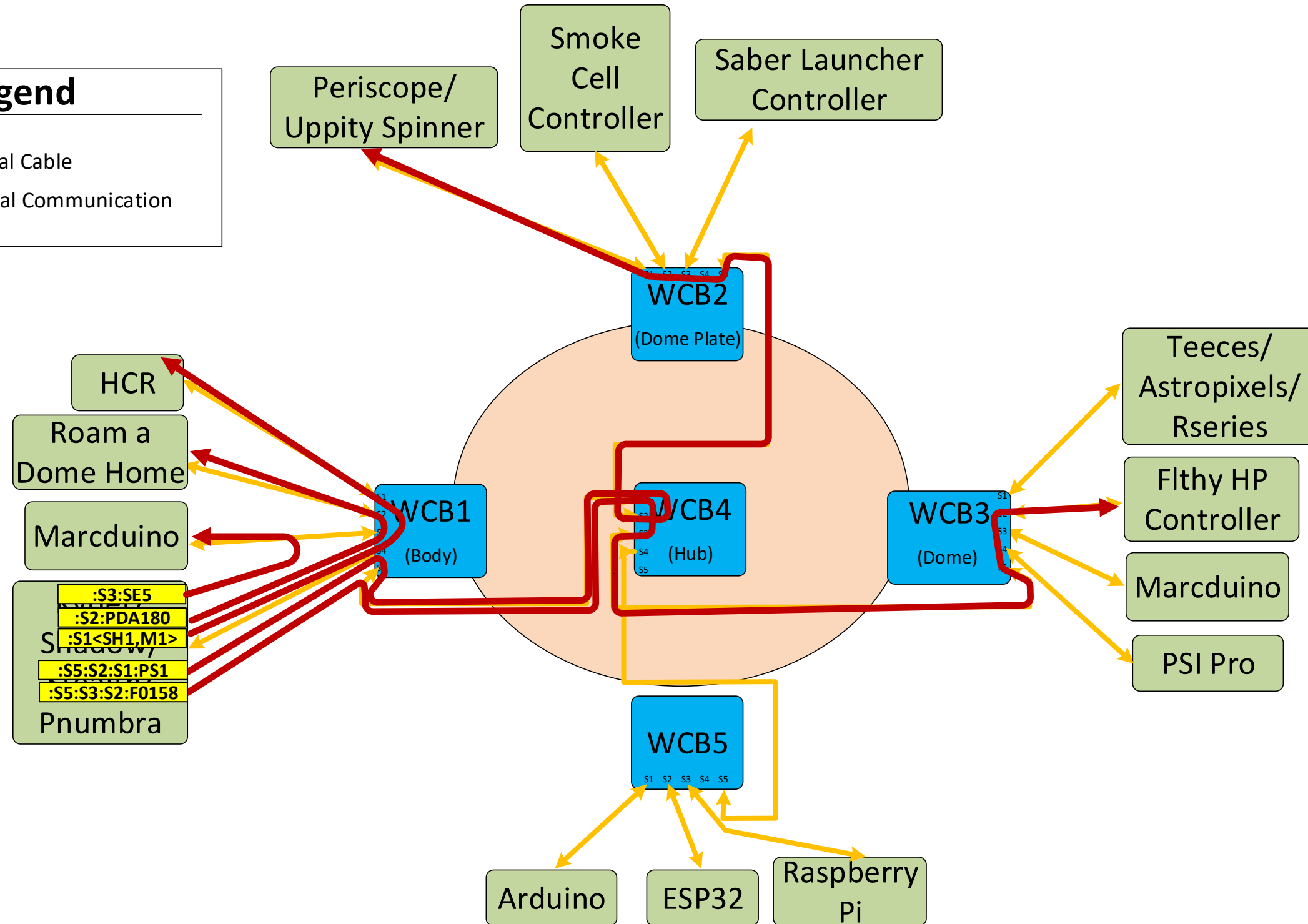
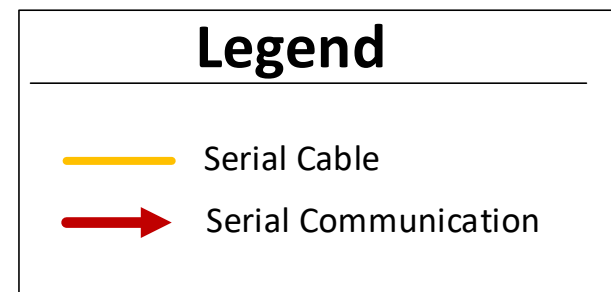


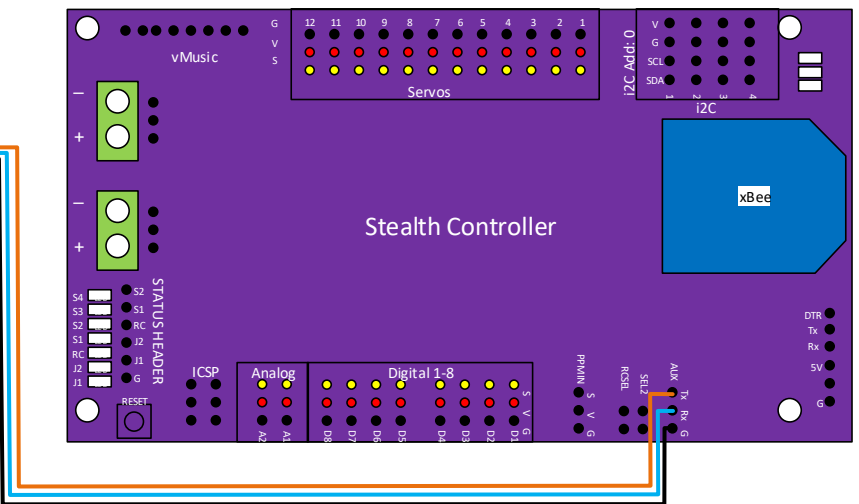
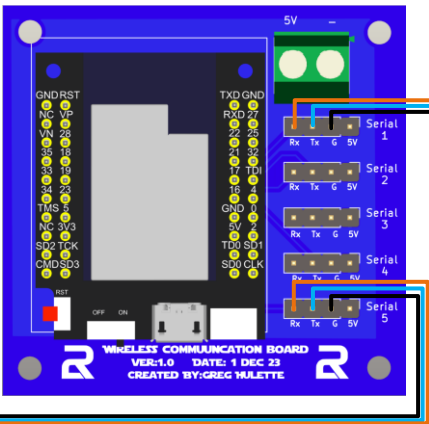
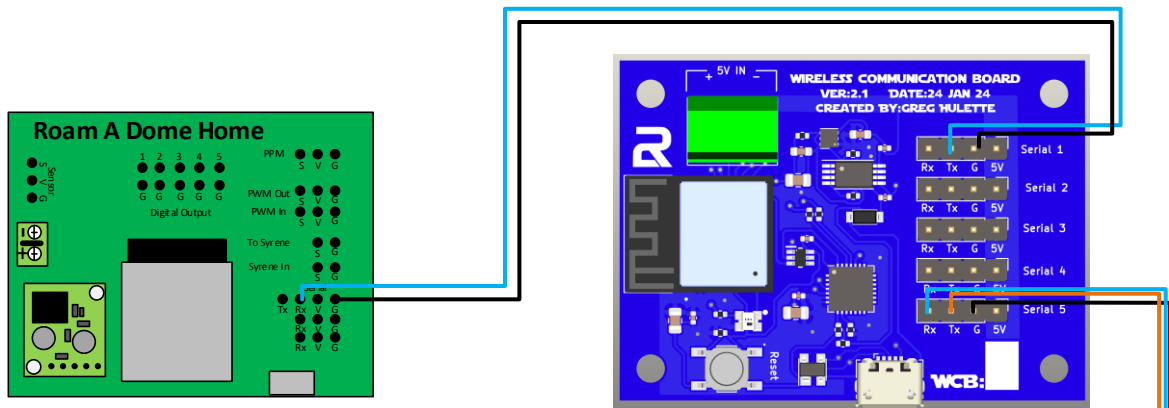
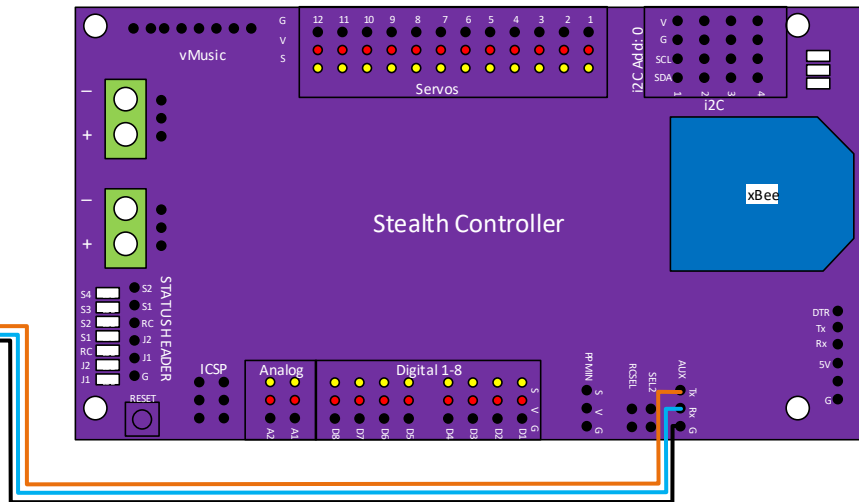
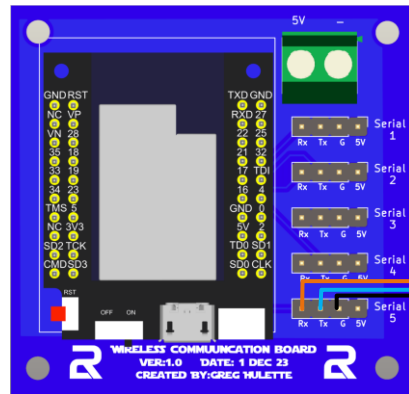
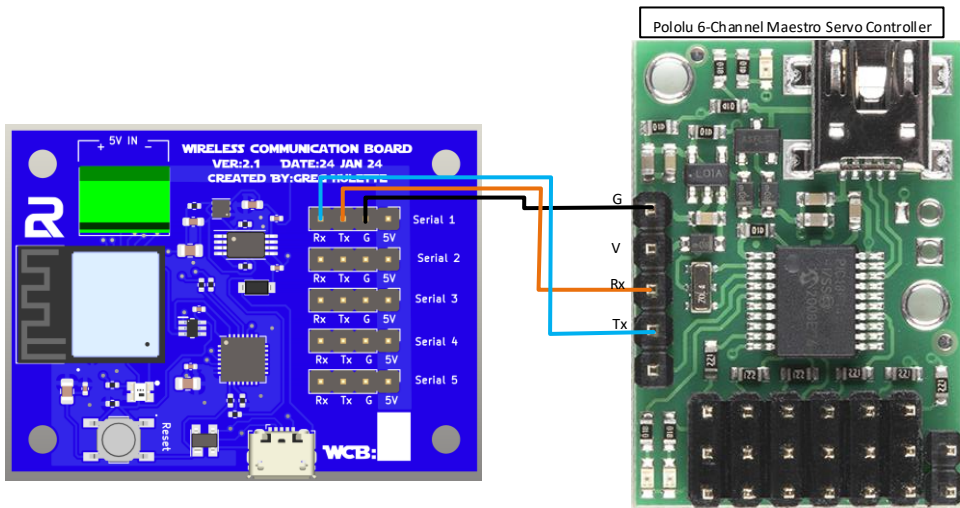
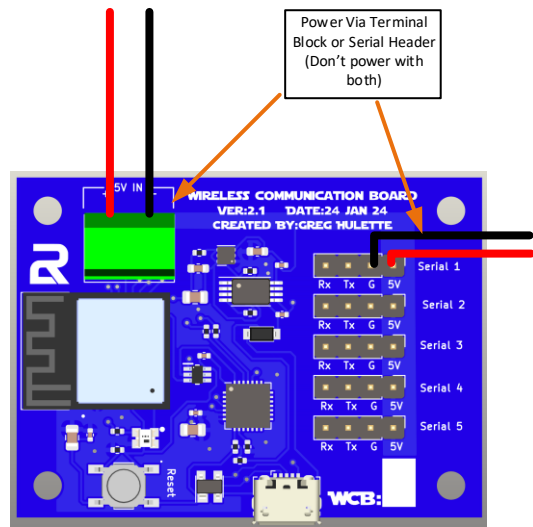
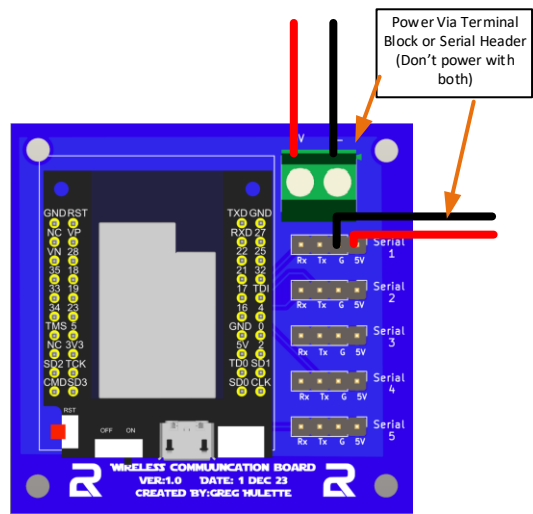
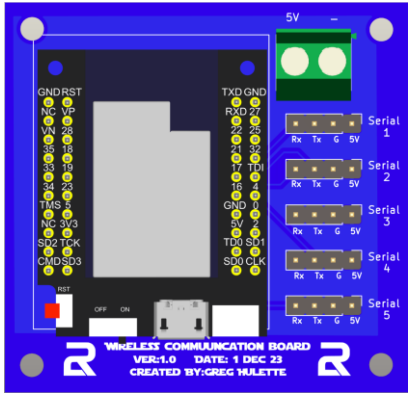
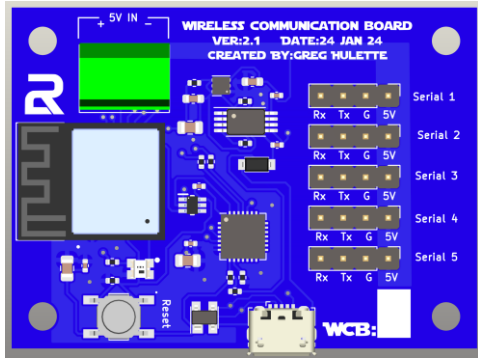


**Option 1
(Passthrough)**

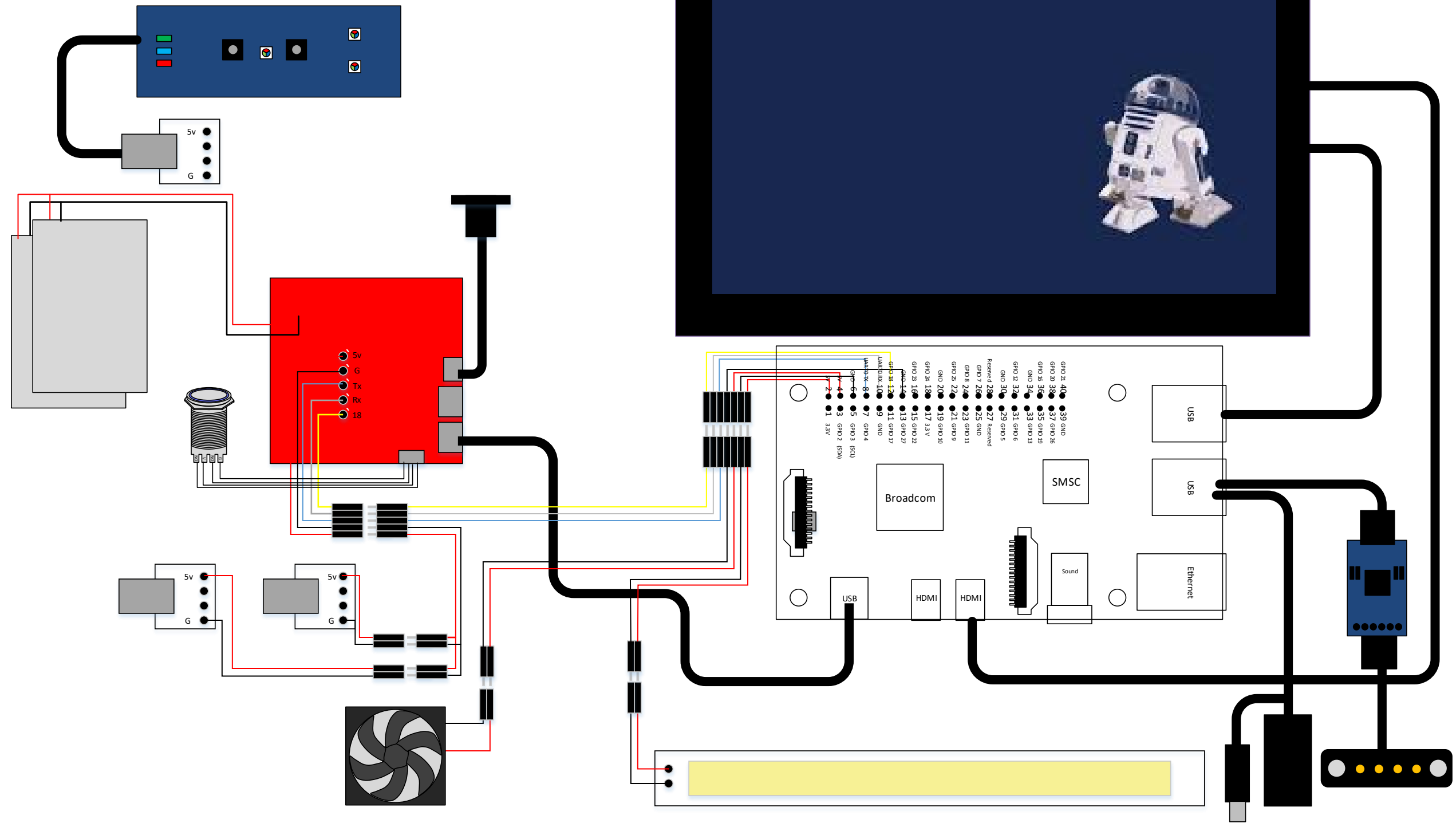


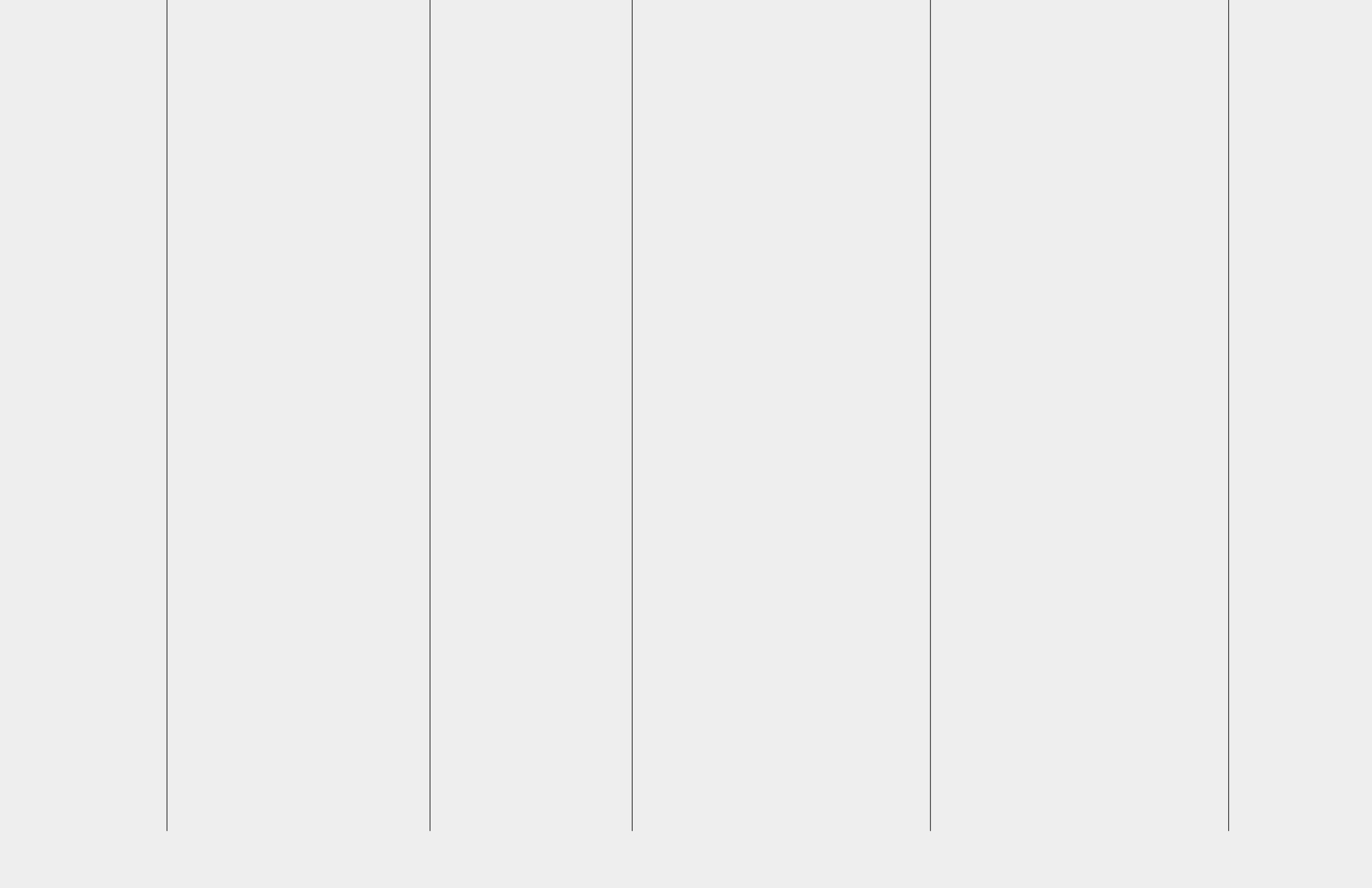
Option 2 (Hub and Spoke)





Needs to be verified and Updated







Back to Quick Reference

Action ^{Tab}

Gesture

Stealth Config.txt

Space Holer

**Body Servo
Expander
I2C = 9**

**Body Arduino Pro
Mini
I2C = 8**

Body Arduino Mega
I2C = 7

Dome Expander
I2C = 10

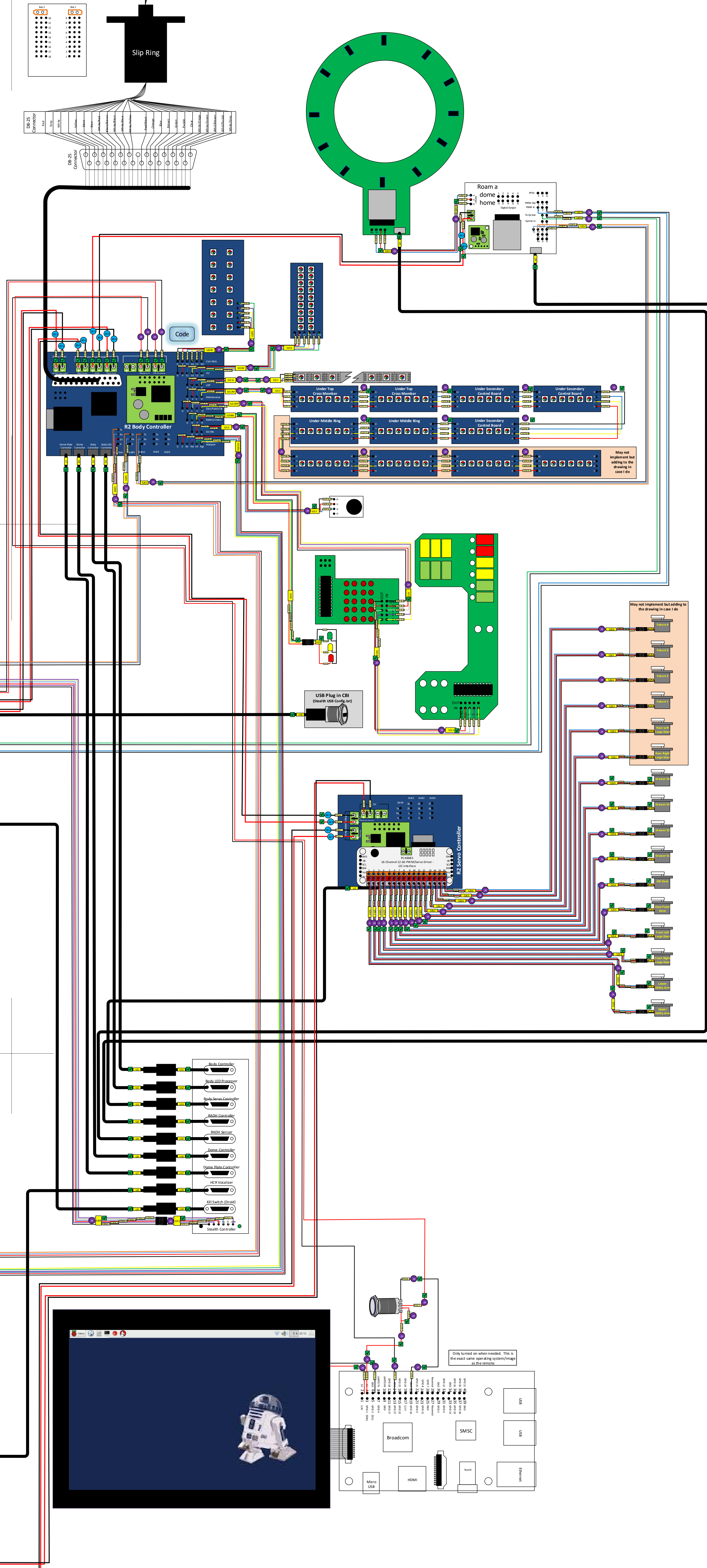
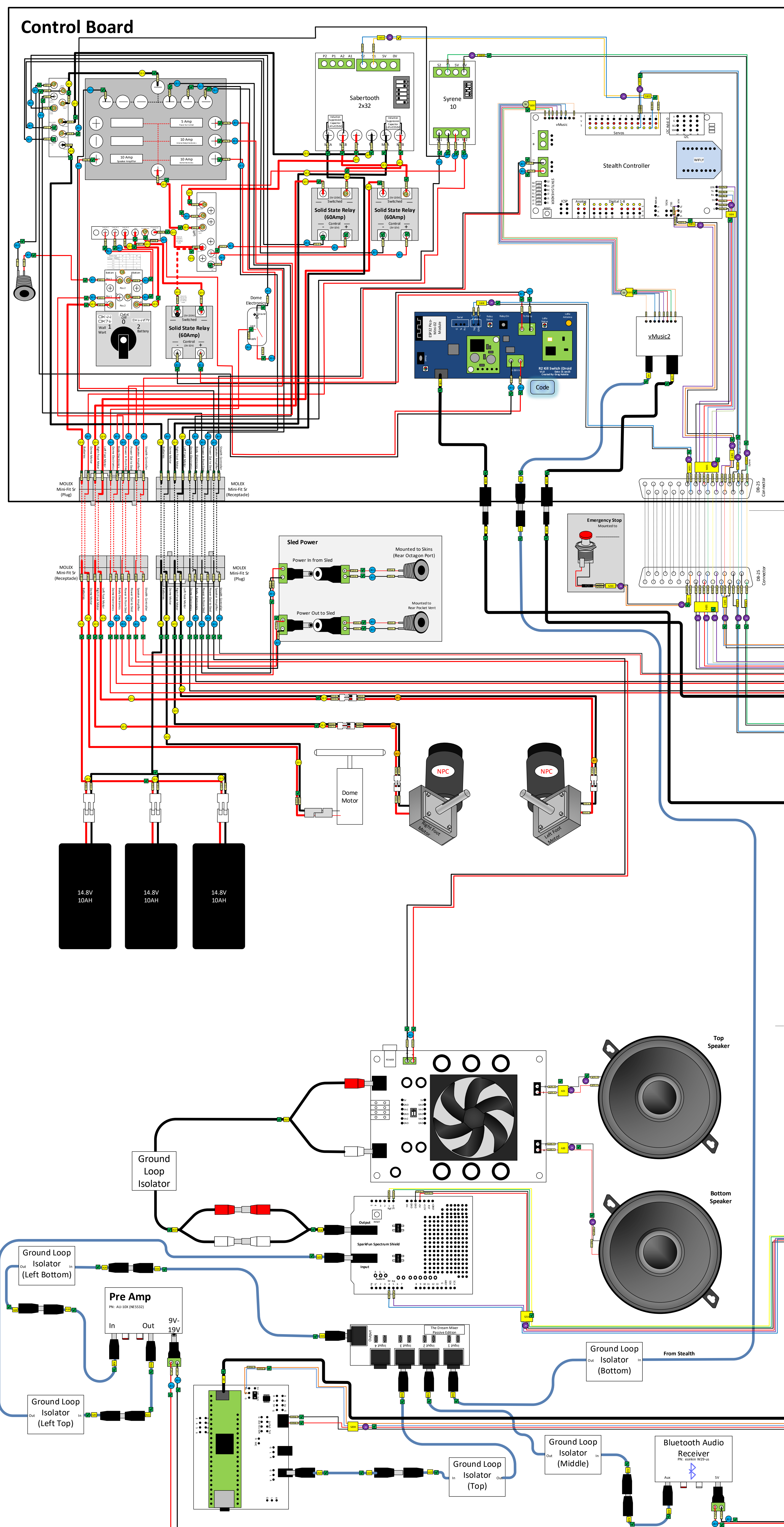
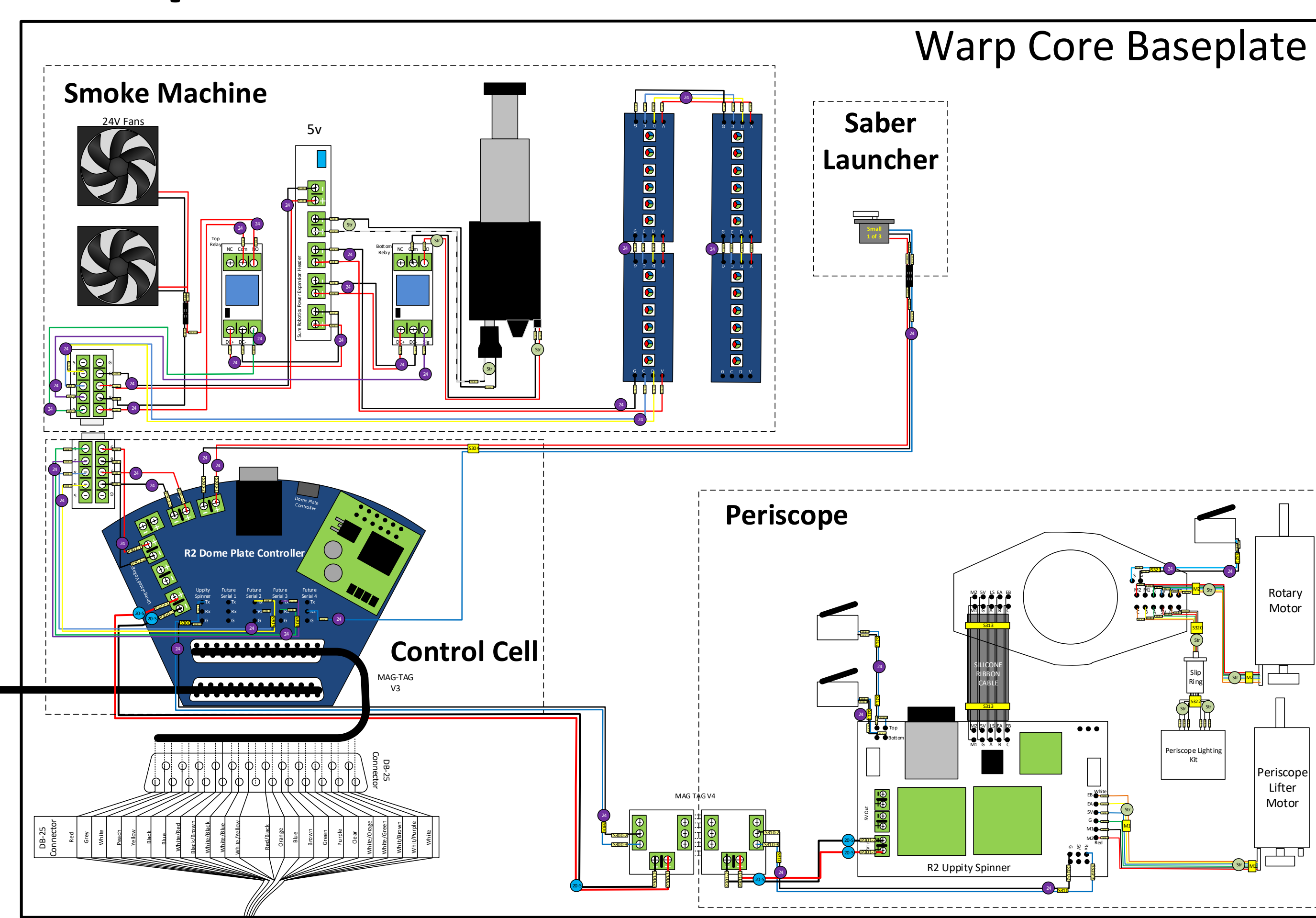
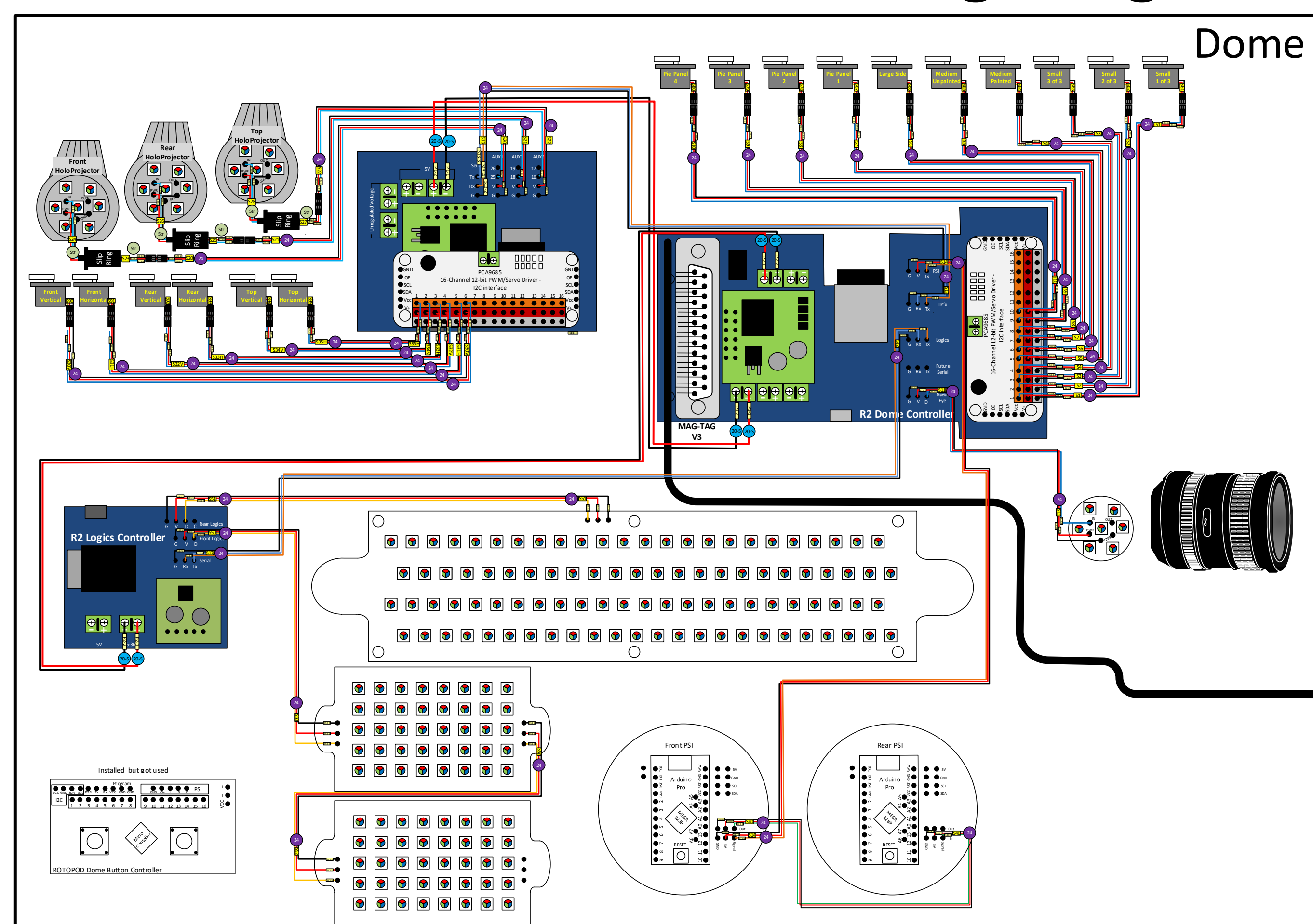
Flthy HP
I2C = 25

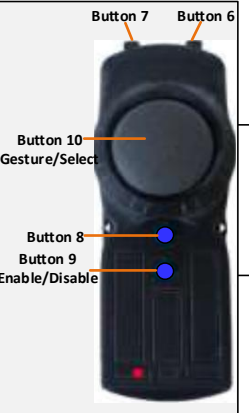
HP Servo
Board
I2C = 65

Teeces (Serial)

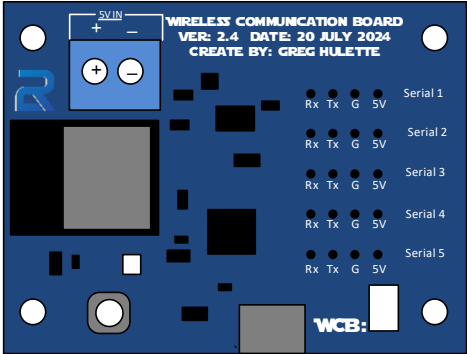
Teeces (Serial)

R2D2-Wiring-Diagram--05-Sep-2026-10-31-PM

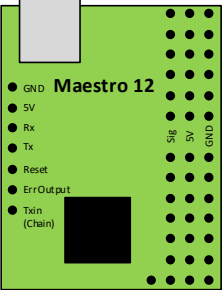
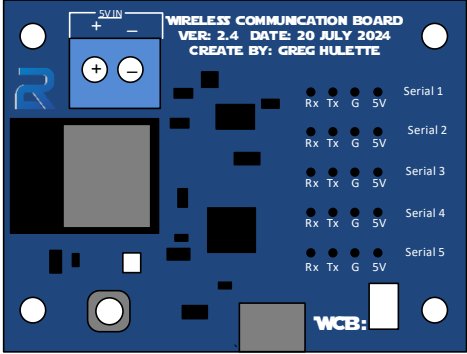


Actions	Details	Actions	Details	Actions	Details	Actions	Details
Normal Operations	g=4,5,1 a=A01	Harlem Shake	g=454,5,5 a=A06	Open/Close Easing	g=45454,5,9 a=A13	Toggle Lights Button 1  Vader Theme LONG BTN 1	b=1,5,13 a=A10 lb=1,5,11 a=A16
Panel Wave	g=6,5,2 a=A02	Door Wave	g=656,5,6 a=A12			Fart Noise Button 2  Star Wars Theme LONG BTN 2	b=2,5,14 a=A21 lb=2,5,10 a=A15
Panel Wave Fast	g=2,5,3 a=A03	Short Circuit	g=252,5,7 a=A14			Flutter Button 3  Wave Utility Arm LONG BTN 3	b=3,5,15 a=A08 lb=3,5,17 a=A22
Dome & Periscope	g=8,5,4 a=A04	Periscope Seq 2	g=858,5,8 a=A17	Periscope Seq 10	g=85858,5,12 a=A18	Toggle Door Button 4  Open/Close Arm Saber LONG BTN 4	b=4,5,16 a=A09 lb=4,5,18 a=A023
						Sound Chat  Launch Saber	b=6,1,2 lb=6,5,19 a=A024
						Sound Bleep  Smoke Sequence	b=7,1,5 lb=7,5,20 a=A025
						Sound Scream 	b=8,1,6 lb=8,5,21 a=A026
						Sound 	b=9,1,2 lb=9,5,22 a=A027
							

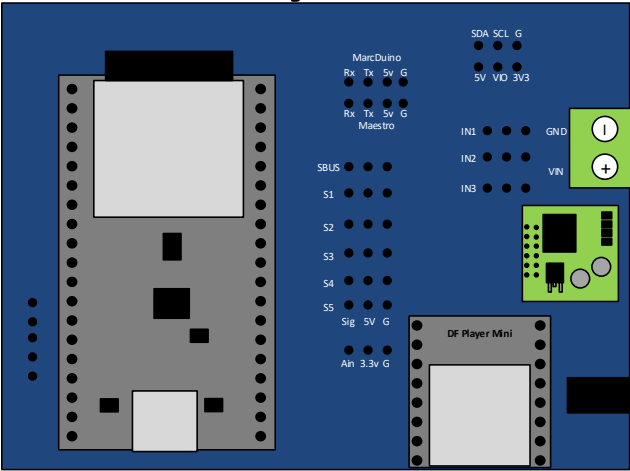
WCB2



WCB2



Kyber



WCB1

